TERS LAPLACE DÖNÜŞÜM FORMÜLLERİ

$$\mathcal{L}^{-1}[F(s)] = f(t)$$

$$f(t) = \frac{1}{2\pi j}\int_{\alpha-j\infty}^{\alpha-j\infty} e^{s^t}\ F(s)ds$$

Katlı Kök Durumu

$$F(s) = \frac{N(s)}{D(s)} = \frac{C_q}{(s-p_r)^q} + \frac{C_{q-1}}{(s-p_r)^{q-1}} + \cdots + \frac{C_1}{(s-p_r)}$$

$$C_q = \lim_{s\to p_r}[(s-p_r)^q \cdot F(s)]$$

$$C_{q-1} = \lim_{s\to p_r} \frac{d}{ds}[(s-p_r)^q \cdot F(s)]$$

$$C_{q-i} = \lim_{s\to p_r} \frac{1}{i!}\frac{d^i}{ds^i}\ [(s-p_r)^q \cdot F(s)]$$

$$f(t) = \left[\frac{C_q}{(q-1)!}\ t^{q-1} + \frac{C_{q-1}}{(q-2)!}t^{q-2} + \cdots\right] e^{p_r \cdot t}$$

Kompleks Eşlenik Kök Durumu

$$F(s) = \frac{\alpha_1 s+\alpha_2}{(s-p_1)(s-p_2)} \qquad \begin{matrix} p_1 = a+jb \\ p_2 = a-jb \end{matrix}$$

$$f(t) = \alpha_1 \cdot e^{a\cdot t} \cdot \cos(bt) + \left(\frac{\alpha_2 + \alpha_1 \cdot a}{b}\right) \cdot e^{a\cdot t} \cdot \sin(bt)$$

Farklı Gerçek Kök Durumu

$$F(s) = \frac{A_1}{s-p_1} + \frac{A_2}{s-p_2} + \cdots$$

$$A_k = \lim_{s\to p_k}[(s-p_k)F(s)]$$

$$f(t) = A_1 \cdot e^{p_1 t} + A_2 \cdot e^{p_2 t} + \cdots$$

Appendix A

TABLE OF GENERAL PROPERTIES OF LAPLACE TRANSFORMS

$$f(s) = \int_0^\infty e^{-st} F(t)\, dt$$

	$f(s)$	$F(t)$
1.	$a f_1(s) + b f_2(s)$	$a F_1(t) + b F_2(t)$
2.	$f(s/a)$	$a F(at)$
3.	$f(s-a)$	$e^{at} F(t)$
4.	$e^{-as} f(s)$	$u(t-a) = \begin{cases} F(t-a) & t > a \\ 0 & t < a \end{cases}$
5.	$s f(s) - F(0)$	$F'(t)$
6.	$s^2 f(s) - s F(0) - F'(0)$	$F''(t)$
7.	$s^n f(s) - s^{n-1} F(0) - s^{n-2} F'(0) - \cdots - F^{(n-1)}(0)$	$F^{(n)}(t)$
8.	$f'(s)$	$-t F(t)$
9.	$f''(s)$	$t^2 F(t)$
10.	$f^{(n)}(s)$	$(-1)^n t^n F(t)$
11.	$\frac{f(s)}{s}$	$\int_0^t F(u)\, du$
12.	$\frac{f(s)}{s^n}$	$\int_0^t \cdots \int_0^t F(u)\, du^n = \int_0^t \frac{(t-u)^{n-1}}{(n-1)!} F(u)\, du$
13.	$f(s)\, g(s)$	$\int_0^t F(u)\, G(t-u)\, du$

	$f(s)$	$F(t)$
14.	$\int_s^\infty f(u)\,du$	$\frac{F(t)}{t}$
15.	$\frac{1}{1-e^{-sT}}\int_0^T e^{-su}F(u)\,du$	$F(t) = F(t+T)$
16.	$\frac{f(\sqrt{s})}{s}$	$\frac{1}{\sqrt{\pi t}}\int_0^\infty e^{-u^2/4t}F(u)\,du$
17.	$\frac{1}{s}f(1/s)$	$\int_0^\infty J_0(2\sqrt{ut})\,F(u)\,du$
18.	$\frac{1}{s^{n+1}}f(1/s)$	$t^{n/2}\int_0^\infty u^{-n/2}J_n(2\sqrt{ut})\,F(u)\,du$
19.	$\frac{f(s+1/s)}{s^2+1}$	$\int_0^t J_0(2\sqrt{u(t-u)})\,F(u)\,du$
20.	$\frac{1}{2\sqrt{\pi}}\int_0^\infty u^{-3/2}e^{-s^2/4u}f(u)\,du$	$F(t^2)$
21.	$\frac{f(\ln s)}{s\ln s}$	$\int_0^\infty \frac{t^u F(u)}{\Gamma(u+1)}\,du$
22.	$\frac{P(s)}{Q(s)}$ $P(s)$ = polynomial of degree less than n, $Q(s) = (s-\alpha_1)(s-\alpha_2)\cdots(s-\alpha_n)$ where $\alpha_1, \alpha_2, \ldots, \alpha_n$ are all distinct.	$\sum_{k=1}^{n}\frac{P(\alpha_k)}{Q'(\alpha_k)}e^{\alpha_k t}$

Appendix B

TABLE OF SPECIAL LAPLACE TRANSFORMS

	$f(s)$	$F(t)$
1.	$\frac{1}{s}$	1
2.	$\frac{1}{s^2}$	t
3.	$\frac{1}{s^n}$ $\quad n = 1, 2, 3, \ldots$	$\frac{t^{n-1}}{(n-1)!}$, $\quad 0! = 1$
4.	$\frac{1}{s^n}$ $\quad n > 0$	$\frac{t^{n-1}}{\Gamma(n)}$
5.	$\frac{1}{s-a}$	e^{at}
6.	$\frac{1}{(s-a)^n}$ $\quad n = 1, 2, 3, \ldots$	$\frac{t^{n-1}e^{at}}{(n-1)!}$, $\quad 0! = 1$
7.	$\frac{1}{(s-a)^n}$ $\quad n > 0$	$\frac{t^{n-1}e^{at}}{\Gamma(n)}$
8.	$\frac{1}{s^2+a^2}$	$\frac{\sin at}{a}$
9.	$\frac{s}{s^2+a^2}$	$\cos at$
10.	$\frac{1}{(s-b)^2+a^2}$	$\frac{e^{bt}\sin at}{a}$
11.	$\frac{s-b}{(s-b)^2+a^2}$	$e^{bt}\cos at$
12.	$\frac{1}{s^2-a^2}$	$\frac{\sinh at}{a}$
13.	$\frac{s}{s^2-a^2}$	$\cosh at$
14.	$\frac{1}{(s-b)^2-a^2}$	$\frac{e^{bt}\sinh at}{a}$

	$f(s)$	$F(t)$
15.	$\frac{s-b}{(s-b)^2-a^2}$	$e^{bt}\cosh at$
16.	$\frac{1}{(s-a)(s-b)}$ $a \neq b$	$\frac{e^{bt}-e^{at}}{b-a}$
17.	$\frac{s}{(s-a)(s-b)}$ $a \neq b$	$\frac{be^{bt}-ae^{at}}{b-a}$
18.	$\frac{1}{(s^2+a^2)^2}$	$\frac{\sin at - at\cos at}{2a^3}$
19.	$\frac{s}{(s^2+a^2)^2}$	$\frac{t\sin at}{2a}$
20.	$\frac{s^2}{(s^2+a^2)^2}$	$\frac{\sin at + at\cos at}{2a}$
21.	$\frac{s^3}{(s^2+a^2)^2}$	$\cos at - \frac{1}{2}at\sin at$
22.	$\frac{s^2-a^2}{(s^2+a^2)^2}$	$t\cos at$
23.	$\frac{1}{(s^2-a^2)^2}$	$\frac{at\cosh at - \sinh at}{2a^3}$
24.	$\frac{s}{(s^2-a^2)^2}$	$\frac{t\sinh at}{2a}$
25.	$\frac{s^2}{(s^2-a^2)^2}$	$\frac{\sinh at + at\cosh at}{2a}$
26.	$\frac{s^3}{(s^2-a^2)^2}$	$\cosh at + \frac{1}{2}at\sinh at$
27.	$\frac{s^2+a^2}{(s^2-a^2)^2}$	$t\cosh at$
28.	$\frac{1}{(s^2+a^2)^3}$	$\frac{(3-a^2t^2)\sin at - 3at\cos at}{8a^5}$
29.	$\frac{s}{(s^2+a^2)^3}$	$\frac{t\sin at - at^2\cos at}{8a^3}$
30.	$\frac{s^2}{(s^2+a^2)^3}$	$\frac{(1+a^2t^2)\sin at - at\cos at}{8a^3}$
31.	$\frac{s^3}{(s^2+a^2)^3}$	$\frac{3t\sin at + at^2\cos at}{8a}$

	$f(s)$	$F(t)$
32.	$\frac{s^4}{(s^2+a^2)^3}$	$\frac{(3-a^2t^2)\sin at + 5at\cos at}{8a}$
33.	$\frac{s^5}{(s^2+a^2)^3}$	$\frac{(8-a^2t^2)\cos at - 7at\sin at}{8}$
34.	$\frac{3s^2-a^2}{(s^2+a^2)^3}$	$\frac{t^2\sin at}{2a}$
35.	$\frac{s^3-3a^2s}{(s^2+a^2)^3}$	$\frac{1}{2}t^2\cos at$
36.	$\frac{s^4-6a^2s^2+a^4}{(s^2+a^2)^4}$	$\frac{1}{6}t^3\cos at$
37.	$\frac{s^3-a^2s}{(s^2+a^2)^4}$	$\frac{t^3\sin at}{24a}$
38.	$\frac{1}{(s^2-a^2)^3}$	$\frac{(3+a^2t^2)\sinh at - 3at\cosh at}{8a^5}$
39.	$\frac{s}{(s^2-a^2)^3}$	$\frac{at^2\cosh at - t\sinh at}{8a^3}$
40.	$\frac{s^2}{(s^2-a^2)^3}$	$\frac{at\cosh at + (a^2t^2-1)\sinh at}{8a^3}$
41.	$\frac{s^3}{(s^2-a^2)^3}$	$\frac{3t\sinh at + at^2\cosh at}{8a}$
42.	$\frac{s^4}{(s^2-a^2)^3}$	$\frac{(3+a^2t^2)\sinh at + 5at\cosh at}{8a}$
43.	$\frac{s^5}{(s^2-a^2)^3}$	$\frac{(8+a^2t^2)\cosh at + 7at\sinh at}{8}$
44.	$\frac{3s^2+a^2}{(s^2-a^2)^3}$	$\frac{t^2\sinh at}{2a}$
45.	$\frac{s^3+3a^2s}{(s^2-a^2)^3}$	$\frac{1}{2}t^2\cosh at$
46.	$\frac{s^4+6a^2s^2+a^4}{(s^2-a^2)^4}$	$\frac{1}{6}t^3\cosh at$
47.	$\frac{s^3+a^2s}{(s^2-a^2)^4}$	$\frac{t^3\sinh at}{24a}$
48.	$\frac{1}{s^3+a^3}$	$\frac{e^{at/2}}{3a^2}\left\{\sqrt{3}\sin\frac{\sqrt{3}\,at}{2} - \cos\frac{\sqrt{3}\,at}{2} + e^{-3at/2}\right\}$

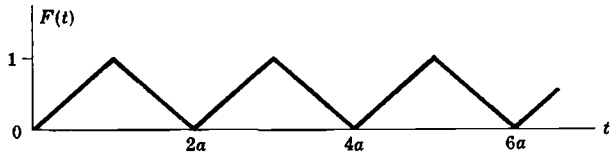
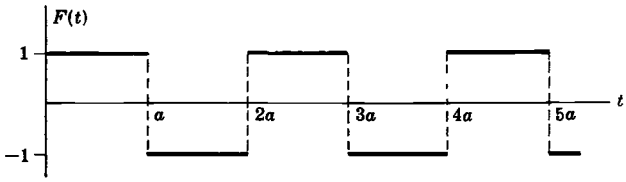
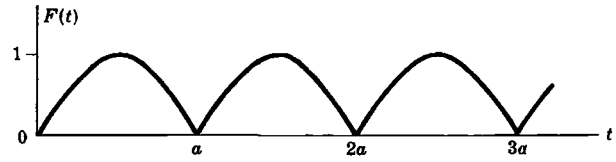
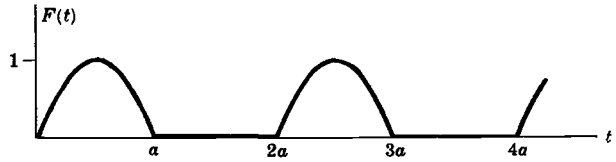
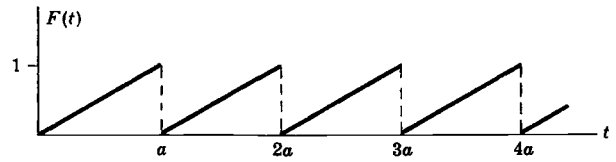
	$f(s)$	$F(t)$
49.	$\frac{s}{s^3 + a^3}$	$\frac{e^{at/2}}{3a}\left\{\cos\frac{\sqrt{3}\,at}{2} + \sqrt{3}\sin\frac{\sqrt{3}\,at}{2} - e^{-3at/2}\right\}$
50.	$\frac{s^2}{s^3 + a^3}$	$\frac{1}{3}\left(e^{-at} + 2e^{at/2}\cos\frac{\sqrt{3}\,at}{2}\right)$
51.	$\frac{1}{s^3 - a^3}$	$\frac{e^{-at/2}}{3a^2}\left\{e^{3at/2} - \cos\frac{\sqrt{3}\,at}{2} - \sqrt{3}\sin\frac{\sqrt{3}\,at}{2}\right\}$
52.	$\frac{s}{s^3 - a^3}$	$\frac{e^{-at/2}}{3a}\left\{\sqrt{3}\sin\frac{\sqrt{3}\,at}{2} - \cos\frac{\sqrt{3}\,at}{2} + e^{3at/2}\right\}$
53.	$\frac{s^2}{s^3 - a^3}$	$\frac{1}{3}\left(e^{at} + 2e^{-at/2}\cos\frac{\sqrt{3}\,at}{2}\right)$
54.	$\frac{1}{s^4 + 4a^4}$	$\frac{1}{4a^3}(\sin at\cosh at - \cos at\sinh at)$
55.	$\frac{s}{s^4 + 4a^4}$	$\frac{\sin at\sinh at}{2a^2}$
56.	$\frac{s^2}{s^4 + 4a^4}$	$\frac{1}{2a}(\sin at\cosh at + \cos at\sinh at)$
57.	$\frac{s^3}{s^4 + 4a^4}$	$\cos at\cosh at$
58.	$\frac{1}{s^4 - a^4}$	$\frac{1}{2a^3}(\sinh at - \sin at)$
59.	$\frac{s}{s^4 - a^4}$	$\frac{1}{2a^2}(\cosh at - \cos at)$
60.	$\frac{s^2}{s^4 - a^4}$	$\frac{1}{2a}(\sinh at + \sin at)$
61.	$\frac{s^3}{s^4 - a^4}$	$\frac{1}{2}(\cosh at + \cos at)$
62.	$\frac{1}{\sqrt{s+a} + \sqrt{s+b}}$	$\frac{e^{-bt} - e^{-at}}{2(b-a)\sqrt{\pi t^3}}$
63.	$\frac{1}{s\sqrt{s+a}}$	$\frac{\operatorname{erf}\sqrt{at}}{\sqrt{a}}$
64.	$\frac{1}{\sqrt{s}\,(s-a)}$	$\frac{e^{at}\operatorname{erf}\sqrt{at}}{\sqrt{a}}$
65.	$\frac{1}{\sqrt{s-a} + b}$	$e^{at}\left\{\frac{1}{\sqrt{\pi t}} - b\,e^{b^2t}\operatorname{erfc}(b\sqrt{t})\right\}$

	$f(s)$	$F(t)$
66.	$\dfrac{1}{\sqrt{s^2+a^2}}$	$J_0(at)$
67.	$\dfrac{1}{\sqrt{s^2-a^2}}$	$I_0(at)$
68.	$\dfrac{(\sqrt{s^2+a^2}-s)^n}{\sqrt{s^2+a^2}} \quad n>-1$	$a^n J_n(at)$
69.	$\dfrac{(s-\sqrt{s^2-a^2})^n}{\sqrt{s^2-a^2}} \quad n>-1$	$a^n I_n(at)$
70.	$\dfrac{e^{b(s-\sqrt{s^2+a^2})}}{\sqrt{s^2+a^2}}$	$J_0(a\sqrt{t(t+2b)})$
71.	$\dfrac{e^{-b\sqrt{s^2+a^2}}}{\sqrt{s^2+a^2}}$	$\begin{cases} J_0(a\sqrt{t^2-b^2}) & t>b \\ 0 & t<b \end{cases}$
72.	$\dfrac{1}{(s^2+a^2)^{3/2}}$	$\dfrac{tJ_1(at)}{a}$
73.	$\dfrac{s}{(s^2+a^2)^{3/2}}$	$tJ_0(at)$
74.	$\dfrac{s^2}{(s^2+a^2)^{3/2}}$	$J_0(at)-atJ_1(at)$
75.	$\dfrac{1}{(s^2-a^2)^{3/2}}$	$\dfrac{tI_1(at)}{a}$
76.	$\dfrac{s}{(s^2-a^2)^{3/2}}$	$tI_0(at)$
77.	$\dfrac{s^2}{(s^2-a^2)^{3/2}}$	$I_0(at)+atI_1(at)$
78.	$\dfrac{1}{s(e^s-1)}=\dfrac{e^{-s}}{s(1-e^{-s})}$ See also entry 141, Page 254.	$F(t)=n, \quad n\leqq t<n+1, \quad n=0,1,2,\ldots$
79.	$\dfrac{1}{s(e^s-r)}=\dfrac{e^{-s}}{s(1-re^{-s})}$	$F(t)=\sum_{k=1}^{[t]} r^k$ where $[t]$ = greatest integer $\leqq t$
80.	$\dfrac{e^s-1}{s(e^s-r)}=\dfrac{1-e^{-s}}{s(1-re^{-s})}$ See also entry 143, Page 254.	$F(t)=r^n, \quad n\leqq t<n+1, \quad n=0,1,2,\ldots$
81.	$\dfrac{e^{-a/s}}{\sqrt{s}}$	$\dfrac{\cos 2\sqrt{at}}{\sqrt{\pi t}}$

	$f(s)$	$F(t)$
82.	$\frac{e^{-a/s}}{s^{3/2}}$	$\frac{\sin 2\sqrt{at}}{\sqrt{\pi a}}$
83.	$\frac{e^{-a/s}}{s^{n+1}} \qquad n > -1$	$\left(\frac{t}{a}\right)^{n/2} J_n(2\sqrt{at})$
84.	$\frac{e^{-a\sqrt{s}}}{\sqrt{s}}$	$\frac{e^{-a^2/4t}}{\sqrt{\pi t}}$
85.	$e^{-a\sqrt{s}}$	$\frac{a}{2\sqrt{\pi t^3}} e^{-a^2/4t}$
86.	$\frac{1 - e^{-a\sqrt{s}}}{s}$	$\operatorname{erf}(a/2\sqrt{t})$
87.	$\frac{e^{-a\sqrt{s}}}{s}$	$\operatorname{erfc}(a/2\sqrt{t})$
88.	$\frac{e^{-a\sqrt{s}}}{\sqrt{s}(\sqrt{s}+b)}$	$e^{b(bt+a)} \operatorname{erfc}\left(b\sqrt{t} + \frac{a}{2\sqrt{t}}\right)$
89.	$\frac{e^{-a/\sqrt{s}}}{s^{n+1}} \qquad n > -1$	$\frac{1}{\sqrt{\pi t}\, a^{2n+1}} \int_0^\infty u^n e^{-u^2/4a^2t} J_{2n}(2\sqrt{u})\, du$
90.	$\ln\left(\frac{s+a}{s+b}\right)$	$\frac{e^{-bt} - e^{-at}}{t}$
91.	$\frac{\ln[(s^2 + a^2)/a^2]}{2s}$	$\operatorname{Ci}(at)$
92.	$\frac{\ln[(s + a)/a]}{s}$	$\operatorname{Ei}(at)$
93.	$-\frac{(\gamma + \ln s)}{s}$ γ = Euler's constant = .5772156...	$\ln t$
94.	$\ln\left(\frac{s^2 + a^2}{s^2 + b^2}\right)$	$\frac{2(\cos at - \cos bt)}{t}$
95.	$\frac{\pi^2}{6s} + \frac{(\gamma + \ln s)^2}{s}$ γ = Euler's constant = .5772156...	$\ln^2 t$
96.	$\frac{\ln s}{s}$	$-(\ln t + \gamma)$ γ = Euler's constant = .5772156...
97.	$\frac{\ln^2 s}{s}$	$(\ln t + \gamma)^2 - \frac{1}{6}\pi^2$ γ = Euler's constant = .5772156...

	$f(s)$	$F(t)$
98.	$\dfrac{\Gamma'(n+1) - \Gamma(n+1)\ln s}{s^{n+1}} \qquad n > -1$	$t^n \ln t$
99.	$\tan^{-1}(a/s)$	$\dfrac{\sin at}{t}$
100.	$\dfrac{\tan^{-1}(a/s)}{s}$	$\mathrm{Si}(at)$
101.	$\dfrac{e^{a/s}}{\sqrt{s}}\operatorname{erfc}(\sqrt{a/s})$	$\dfrac{e^{-2\sqrt{at}}}{\sqrt{\pi t}}$
102.	$e^{s^2/4a^2}\operatorname{erfc}(s/2a)$	$\dfrac{2a}{\sqrt{\pi}}e^{-a^2t^2}$
103.	$\dfrac{e^{s^2/4a^2}\operatorname{erfc}(s/2a)}{s}$	$\operatorname{erf}(at)$
104.	$\dfrac{e^{as}\operatorname{erfc}\sqrt{as}}{\sqrt{s}}$	$\dfrac{1}{\sqrt{\pi(t+a)}}$
105.	$e^{as}\,\mathrm{Ei}(as)$	$\dfrac{1}{t+a}$
106.	$\dfrac{1}{a}\left[\cos as\left\{\dfrac{\pi}{2} - \mathrm{Si}(as)\right\} - \sin as\,\mathrm{Ci}(as)\right]$	$\dfrac{1}{t^2+a^2}$
107.	$\sin as\left\{\dfrac{\pi}{2} - \mathrm{Si}(as)\right\} + \cos as\,\mathrm{Ci}(as)$	$\dfrac{t}{t^2+a^2}$
108.	$\dfrac{\cos as\left\{\dfrac{\pi}{2} - \mathrm{Si}(as)\right\} - \sin as\,\mathrm{Ci}(as)}{s}$	$\tan^{-1}(t/a)$
109.	$\dfrac{\sin as\left\{\dfrac{\pi}{2} - \mathrm{Si}(as)\right\} + \cos as\,\mathrm{Ci}(as)}{s}$	$\dfrac{1}{2}\ln\left(\dfrac{t^2+a^2}{a^2}\right)$
110.	$\left[\dfrac{\pi}{2} - \mathrm{Si}(as)\right]^2 + \mathrm{Ci}^2(as)$	$\dfrac{1}{t}\ln\left(\dfrac{t^2+a^2}{a^2}\right)$
111.	0	$\mathcal{N}(t)$
112.	1	$\delta(t)$
113.	e^{-as}	$\delta(t-a)$
114.	$\dfrac{e^{-as}}{s}$ See also entry 139, Page 254.	$u(t-a)$

	$f(s)$	$F(t)$
115.	$\frac{\sinh sx}{s \sinh sa}$	$\frac{x}{a} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin \frac{n\pi x}{a} \cos \frac{n\pi t}{a}$
116.	$\frac{\sinh sx}{s \cosh sa}$	$\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} \sin \frac{(2n-1)\pi x}{2a} \sin \frac{(2n-1)\pi t}{2a}$
117.	$\frac{\cosh sx}{s \sinh sa}$	$\frac{t}{a} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \cos \frac{n\pi x}{a} \sin \frac{n\pi t}{a}$
118.	$\frac{\cosh sx}{s \cosh sa}$	$1 + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} \cos \frac{(2n-1)\pi x}{2a} \cos \frac{(2n-1)\pi t}{2a}$
119.	$\frac{\sinh sx}{s^2 \sinh sa}$	$\frac{xt}{a} + \frac{2a}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \sin \frac{n\pi x}{a} \sin \frac{n\pi t}{a}$
120.	$\frac{\sinh sx}{s^2 \cosh sa}$	$x + \frac{8a}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^2} \sin \frac{(2n-1)\pi x}{2a} \cos \frac{(2n-1)\pi t}{2a}$
121.	$\frac{\cosh sx}{s^2 \sinh sa}$	$\frac{t^2}{2a} + \frac{2a}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos \frac{n\pi x}{a} \left(1 - \cos \frac{n\pi t}{a}\right)$
122.	$\frac{\cosh sx}{s^2 \cosh sa}$	$t + \frac{8a}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{2a} \sin \frac{(2n-1)\pi t}{2a}$
123.	$\frac{\cosh sx}{s^3 \cosh sa}$	$\frac{1}{2}(t^2 + x^2 - a^2) - \frac{16a^2}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} \cos \frac{(2n-1)\pi x}{2a} \cos \frac{(2n-1)\pi t}{2a}$
124.	$\frac{\sinh x\sqrt{s}}{\sinh a\sqrt{s}}$	$\frac{2\pi}{a^2} \sum_{n=1}^{\infty} (-1)^n n\, e^{-n^2\pi^2 t/a^2} \sin \frac{n\pi x}{a}$
125.	$\frac{\cosh x\sqrt{s}}{\cosh a\sqrt{s}}$	$\frac{\pi}{a^2} \sum_{n=1}^{\infty} (-1)^{n-1} (2n-1)\, e^{-(2n-1)^2\pi^2 t/4a^2} \cos \frac{(2n-1)\pi x}{2a}$
126.	$\frac{\sinh x\sqrt{s}}{\sqrt{s} \cosh a\sqrt{s}}$	$\frac{2}{a} \sum_{n=1}^{\infty} (-1)^{n-1} e^{(2n-1)^2\pi^2 t/4a^2} \sin \frac{(2n-1)\pi x}{2a}$
127.	$\frac{\cosh x\sqrt{s}}{\sqrt{s} \sinh a\sqrt{s}}$	$\frac{1}{a} + \frac{2}{a} \sum_{n=1}^{\infty} (-1)^n e^{-n^2\pi^2 t/a^2} \cos \frac{n\pi x}{a}$
128.	$\frac{\sinh x\sqrt{s}}{s \sinh a\sqrt{s}}$	$\frac{x}{a} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} e^{n^2\pi^2 t/a^2} \sin \frac{n\pi x}{a}$
129.	$\frac{\cosh x\sqrt{s}}{s \cosh a\sqrt{s}}$	$1 + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} e^{(2n-1)^2\pi^2 t/4a^2} \cos \frac{(2n-1)\pi x}{2a}$
130.	$\frac{\sinh x\sqrt{s}}{s^2 \sinh a\sqrt{s}}$	$\frac{xt}{a} + \frac{2a^2}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} (1 - e^{n^2\pi^2 t/a^2}) \sin \frac{n\pi x}{a}$
131.	$\frac{\cosh x\sqrt{s}}{s^2 \cosh a\sqrt{s}}$	$\frac{1}{2}(x^2 - a^2) + t - \frac{16a^2}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} e^{(2n-1)^2\pi^2 t/4a^2} \cos \frac{(2n-1)\pi x}{2a}$

	$f(s)$	$F(t)$
132.	$\dfrac{J_0(ix\sqrt{s})}{s\,J_0(ia\sqrt{s})}$	$1 - 2\sum_{n=1}^{\infty} \dfrac{e^{-\lambda_n^2 t/a^2} J_0(\lambda_n x/a)}{\lambda_n J_1(\lambda_n)}$ where $\lambda_1, \lambda_2, \ldots$ are the positive roots of $J_0(\lambda) = 0$
133.	$\dfrac{J_0(ix\sqrt{s})}{s^2 J_0(ia\sqrt{s})}$	$\frac{1}{4}(x^2 - a^2) + t + 2a^2 \sum_{n=1}^{\infty} \dfrac{e^{-\lambda_n^2 t/a^2} J_0(\lambda_n x/a)}{\lambda_n^3 J_1(\lambda_n)}$ where $\lambda_1, \lambda_2, \ldots$ are the positive roots of $J_0(\lambda) = 0$
134.	$\dfrac{1}{as^2} \tanh\left(\dfrac{as}{2}\right)$	**Triangular wave function** 
135.	$\dfrac{1}{s} \tanh\left(\dfrac{as}{2}\right)$	**Square wave function** 
136.	$\dfrac{\pi a}{a^2 s^2 + \pi^2} \coth\left(\dfrac{as}{2}\right)$	**Rectified sine wave function** 
137.	$\dfrac{\pi a}{(a^2 s^2 + \pi^2)(1 - e^{-as})}$	**Half rectified sine wave function** 
138.	$\dfrac{1}{as^2} - \dfrac{e^{-as}}{s(1 - e^{-as})}$	**Saw tooth wave function** 

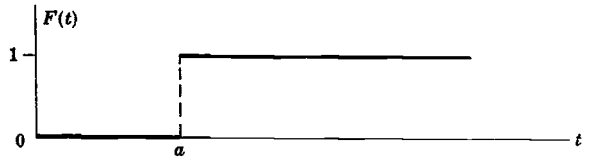
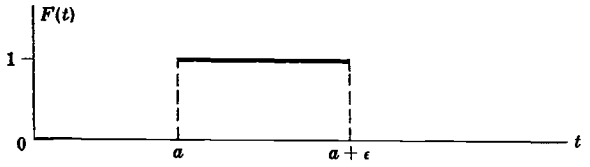
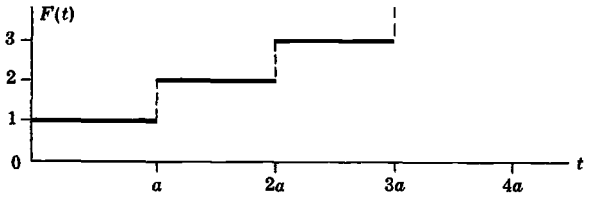
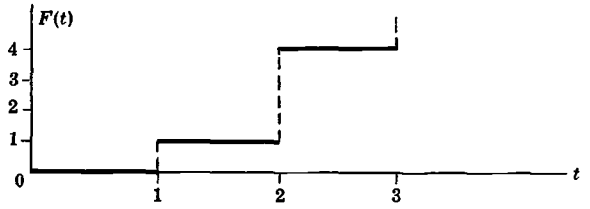
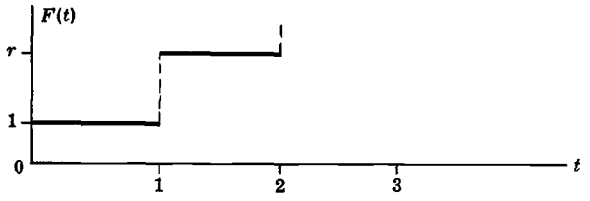
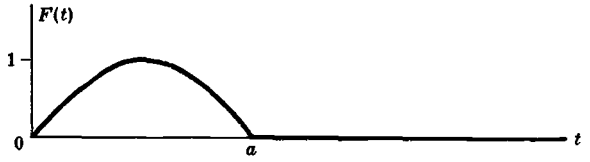
	$f(s)$	$F(t)$
139.	$\frac{e^{-as}}{s}$ See also entry 114, Page 251.	**Heaviside's unit function** $\mathcal{U}(t-a)$ 
140.	$\frac{e^{-as}(1-e^{-\epsilon s})}{s}$	**Pulse function** 
141.	$\frac{1}{s(1-e^{-as})}$ See also entry 78, Page 249.	**Step function** 
142.	$\frac{e^{-s}+e^{-2s}}{s(1-e^{-s})^2}$	$F(t)=n^2, \quad n \leqq t < n+1, \quad n=0,1,2,\ldots$ 
143.	$\frac{1-e^{-s}}{s(1-re^{-s})}$ See also entry 80, Page 249.	$F(t)=r^n, \quad n \leqq t < n+1, \quad n=0,1,2,\ldots$ 
144.	$\frac{\pi a(1+e^{-as})}{a^2s^2+\pi^2}$	$F(t) = \begin{cases} \sin(\pi t/a) & 0 \leqq t \leqq a \\ 0 & t > a \end{cases}$ 

Table of Laplace Transforms

	$f(t)=\mathcal{L}^{-1}\{F(s)\}$	$F(s)=\mathcal{L}\{f(t)\}$		$f(t)=\mathcal{L}^{-1}\{F(s)\}$	$F(s)=\mathcal{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	$\mathbf{e}^{at}$	$\frac{1}{s-a}$
3.	$t^n,\quad n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\quad n=1,2,3,\ldots$	$\frac{1\cdot3\cdot5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$\mathbf{e}^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$\mathbf{e}^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$\mathbf{e}^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22.	$\mathbf{e}^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n\mathbf{e}^{at},\quad n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{\mathbf{e}^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	$\mathbf{e}^{-cs}$
27.	$u_c(t)f(t-c)$	$\mathbf{e}^{-cs}F(s)$	28.	$u_c(t)g(t)$	$\mathbf{e}^{-cs}\mathcal{L}\{g(t+c)\}$
29.	$\mathbf{e}^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\quad n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T \mathbf{e}^{-st}f(t)\,dt}{1-\mathbf{e}^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$	36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			

Table Notes

1. This list is not a complete listing of Laplace transforms and only contains some of the more commonly used Laplace transforms and formulas.

2. Recall the definition of hyperbolic functions.

$$\cosh(t) = \frac{\mathbf{e}^{t} + \mathbf{e}^{-t}}{2} \qquad \sinh(t) = \frac{\mathbf{e}^{t} - \mathbf{e}^{-t}}{2}$$

3. Be careful when using "normal" trig function vs. hyperbolic functions. The only difference in the formulas is the "+ a^2" for the "normal" trig functions becomes a "- a^2" for the hyperbolic functions!

4. Formula #4 uses the Gamma function which is defined as

$$\Gamma(t) = \int_0^\infty \mathbf{e}^{-x} x^{t-1}\, dx$$

   If n is a positive integer then,

$$\Gamma(n+1) = n!$$

   The Gamma function is an extension of the normal factorial function. Here are a couple of quick facts for the Gamma function

$$\Gamma(p+1) = p\Gamma(p)$$

$$p(p+1)(p+2)\cdots(p+n-1) = \frac{\Gamma(p+n)}{\Gamma(p)}$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$